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Generation of Believable Fake Combinational Logic Circuits for Cyber Deception

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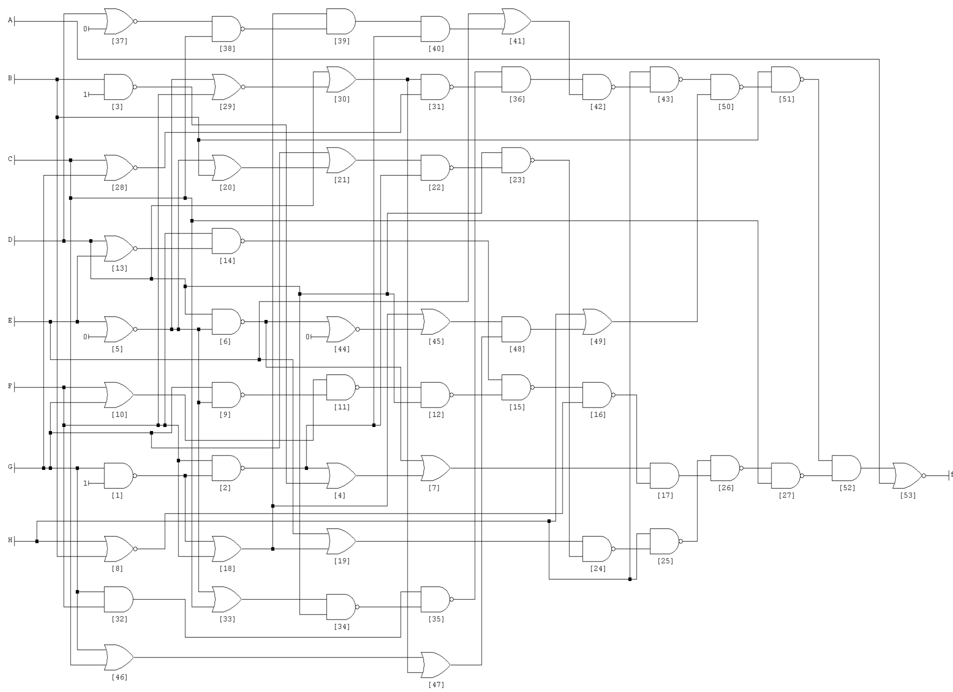
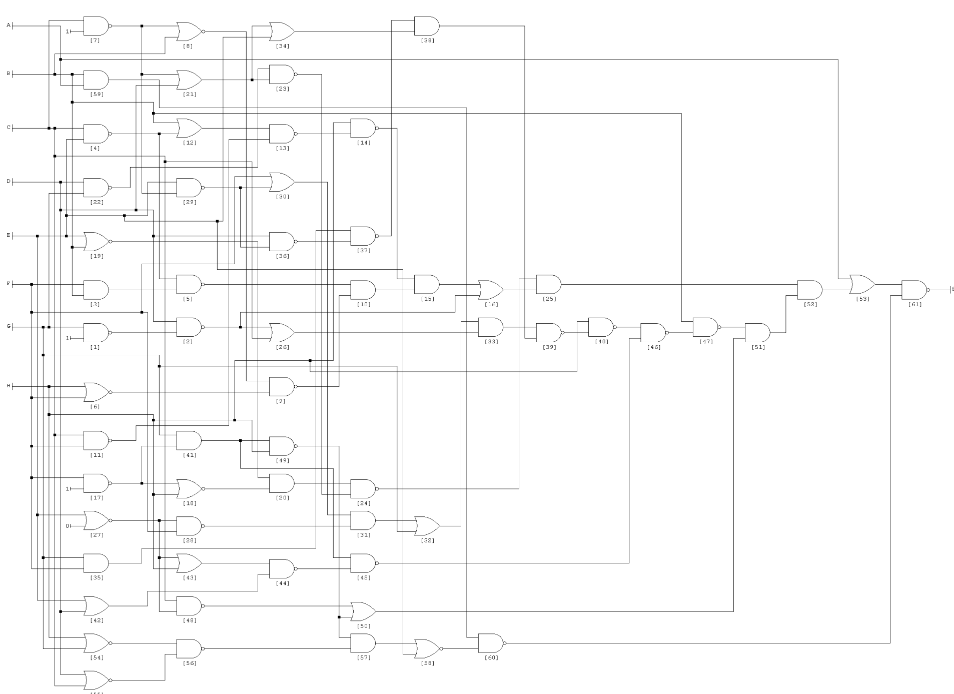
Abstract: The exponential increase in intellectual property (IP) theft has raised concern for organizations and government entities. Existing research suggests the use of fake document for protection against IP theft. To protect the logic circuit present in the IP document, we propose Believable Fake Combinational Logic Circuits Generation Engine (BFCLCGE), which is an extension of our previous work (FBLCGE). We addressed the exhaustive search problem by modifying the original Boolean equation and introducing replacement constraints, thereby reducing overall time complexity. Additionally, we overcame the limitation of FBLCGE by incorporating ESPRESSO algorithm for efficient minimization of Boolean equations with a large number of input variables. To test the believability of generated fake Boolean circuits, we implemented a virtual display (sixteen-segment) and simulated the outputs. The proposed approach is efficient and generates indistinguishable fake circuits, which can be used to create believable fake documents for cyber deception against potential IP theft.

Keywords: Intellectual Property ; Data Ex-filtration ; Combinational Logic Circuits ; Hardware Security ; Cyber Security ; Cyber Deception ; Decoy Logic Circuits ; Decoy Documents

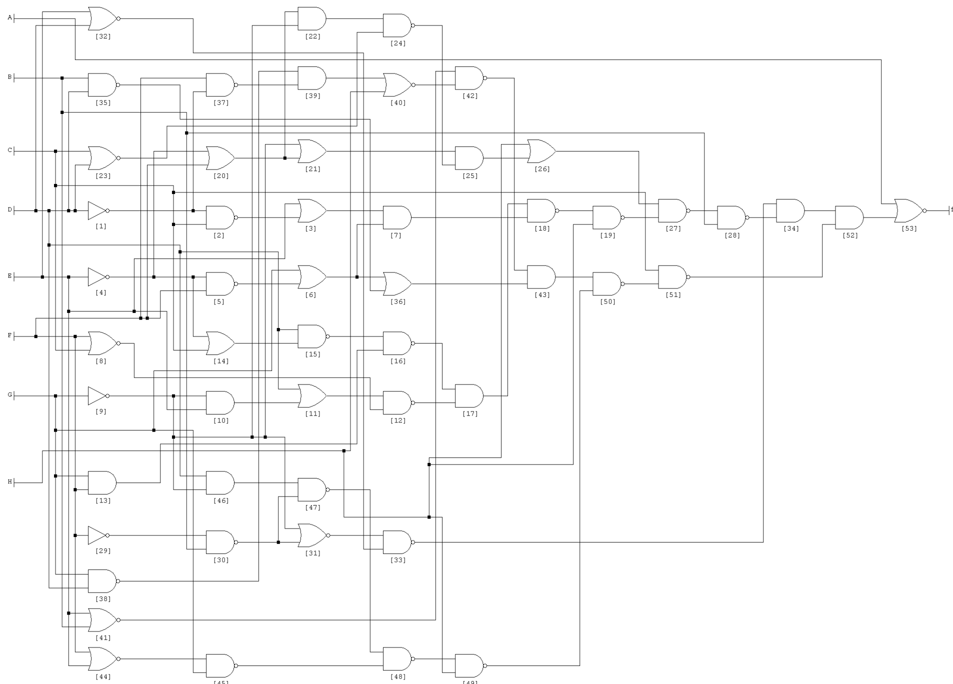
1 Supplementary Information



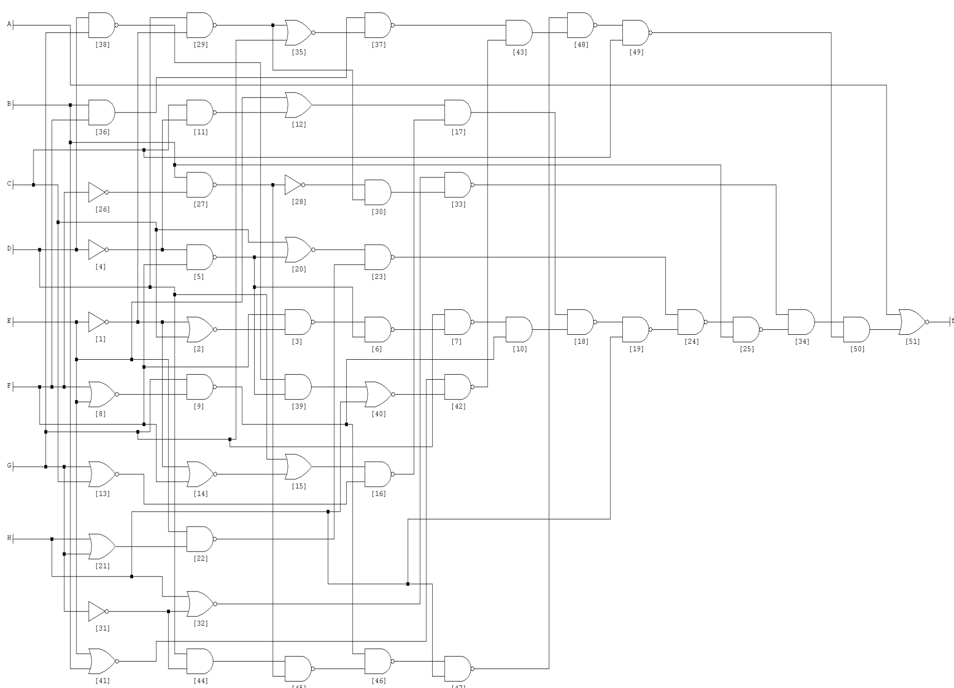
(a) Original: f-segment of sixteen-segment display



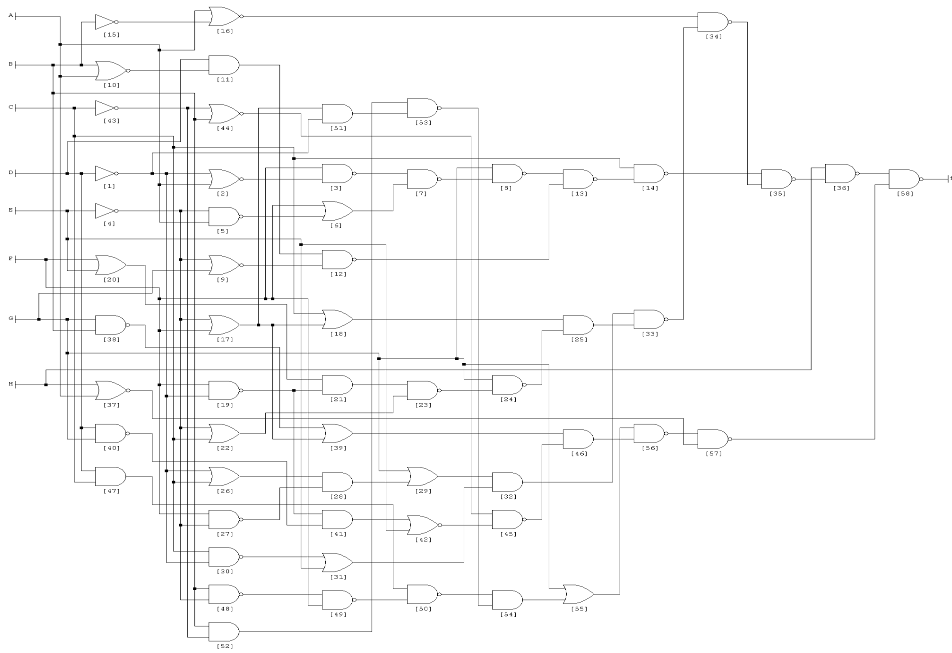
(b) Fake 1: Logic circuit for f-segment of sixteen-segment display



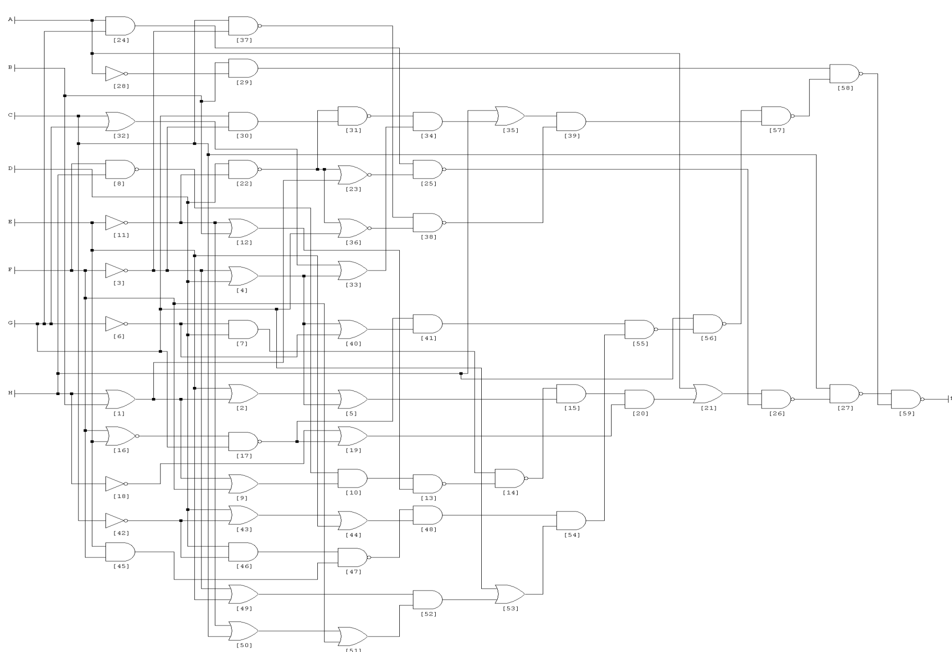
(c) Fake 2: Logic circuit for f-segment of sixteen-segment display



(d) Fake 3: Logic circuit for f-segment of sixteen-segment display

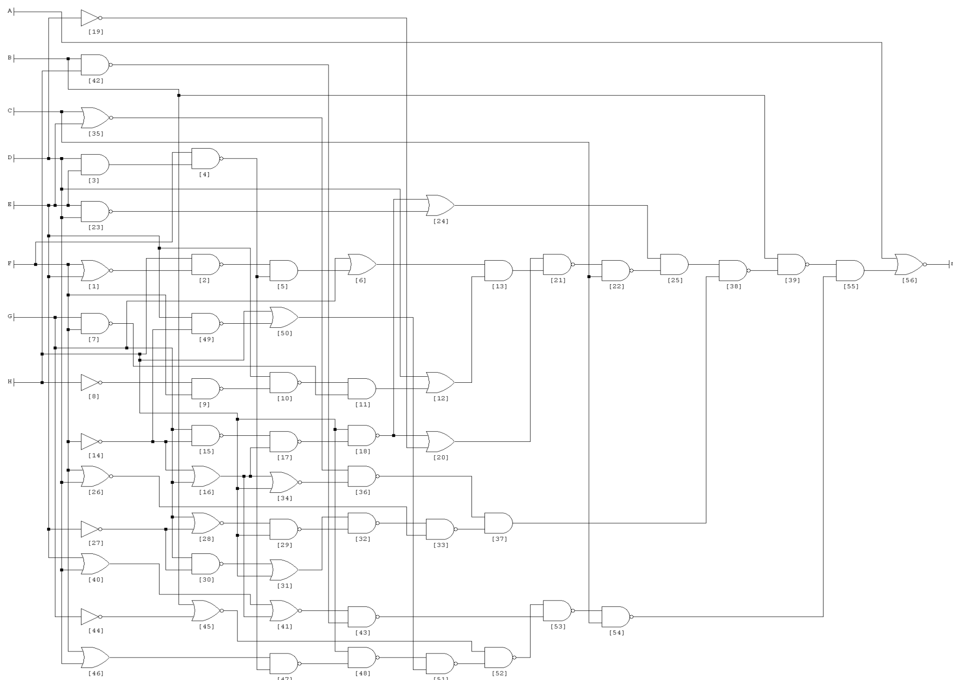


(e) Fake 4: Logic circuit for f-segment of sixteen-segment display

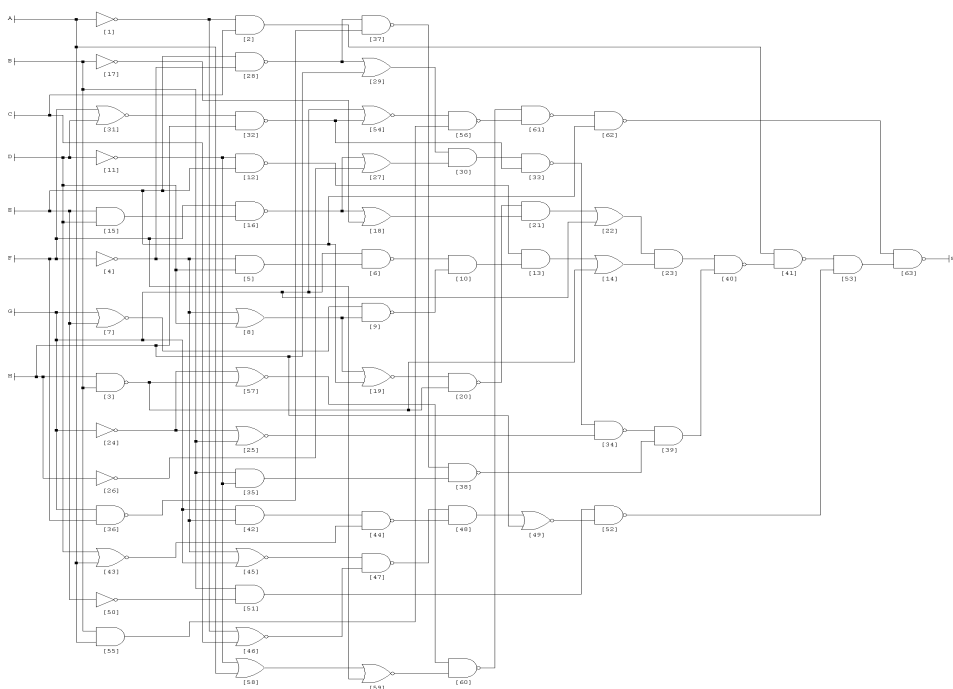


(f) Fake 5: Logic circuit for f-segment of sixteen-segment display

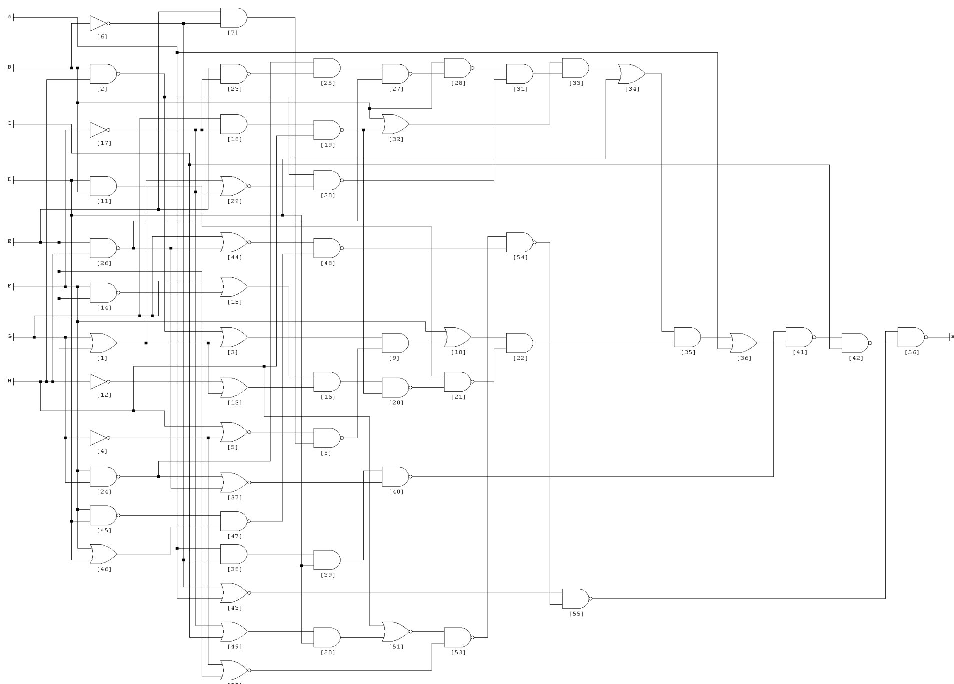
Figure S1: Original and Fake logic circuits for f-segment of sixteen-segment display. (a) represent the circuit diagram for original circuit. The generated fake version of original circuits are depicted in Sub-figures (b), (c), (d), (e), and (f).



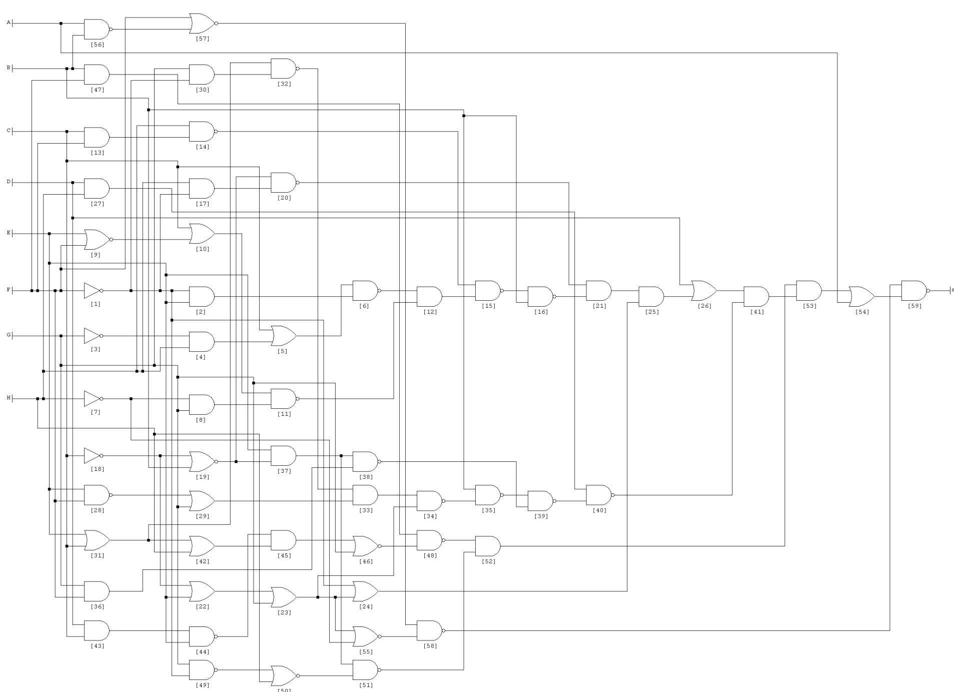
(a) Original: Logic circuit for S-segment of sixteen-segment display



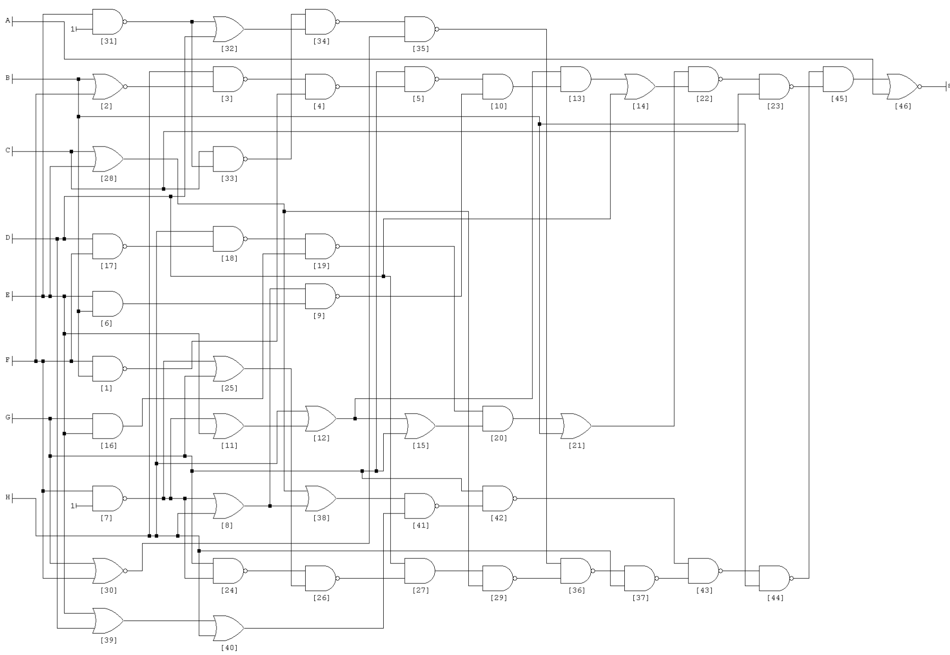
(b) Fake 1: Logic circuit for S-segment of sixteen-segment display



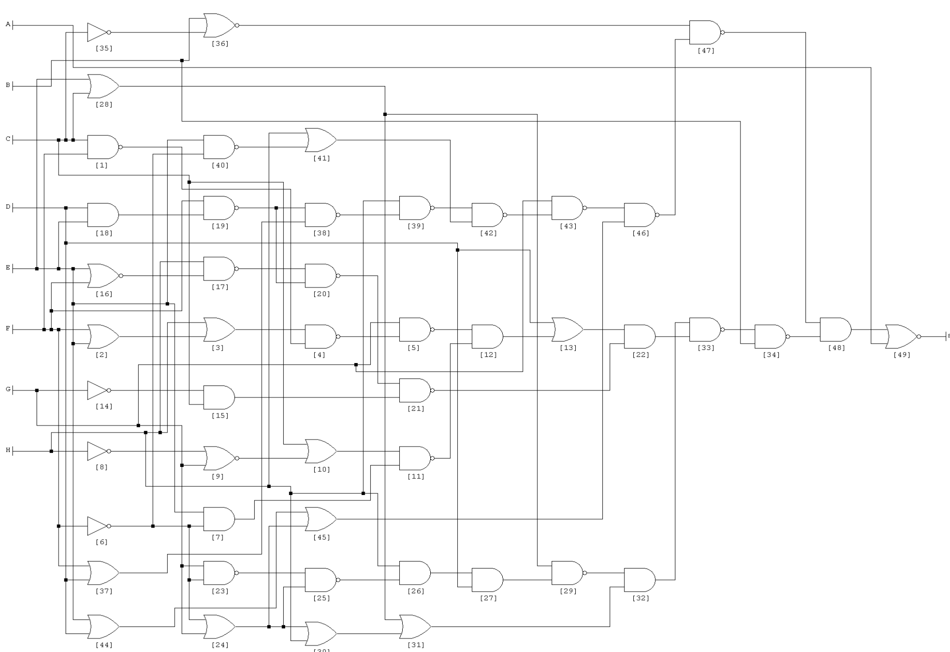
(c) Fake 2: Logic circuit for S-segment of sixteen-segment display



(d) Fake 3: Logic circuit for S-segment of sixteen-segment display

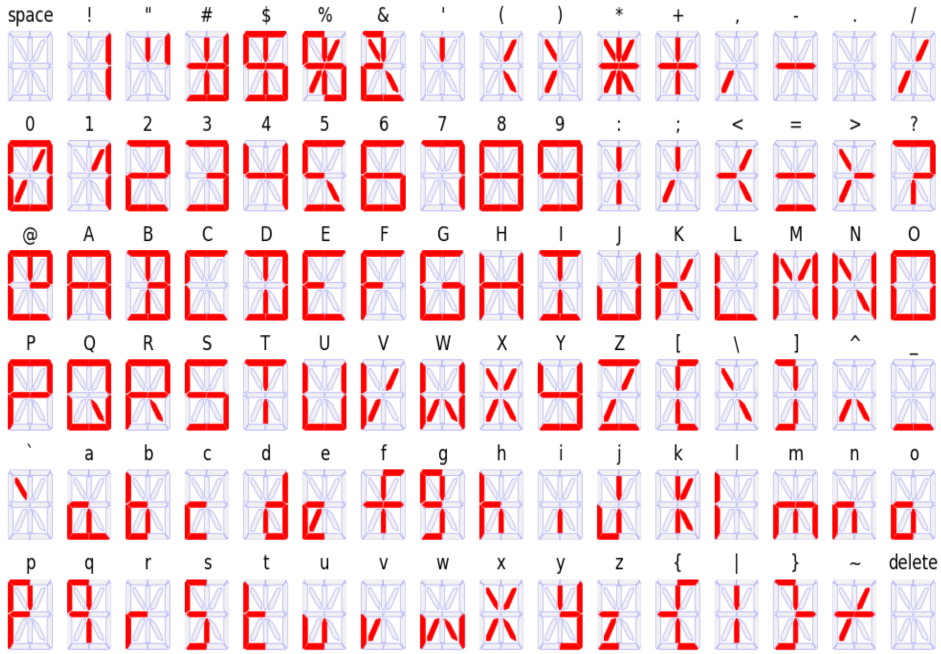


(e) Fake 4: Logic circuit for S-segment of sixteen-segment display

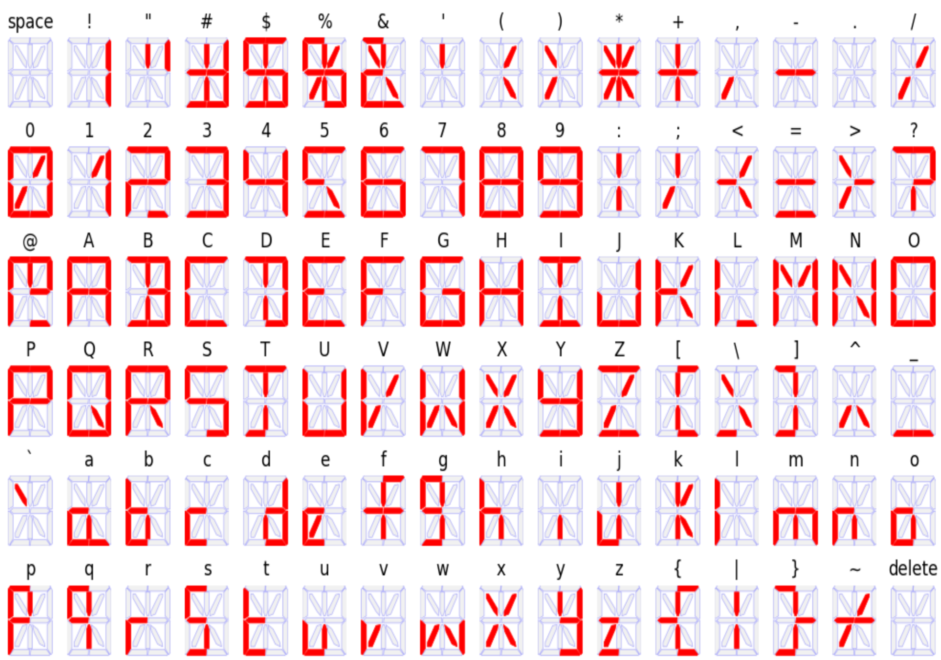


(f) Fake 5: Logic circuit for S-segment of sixteen-segment display

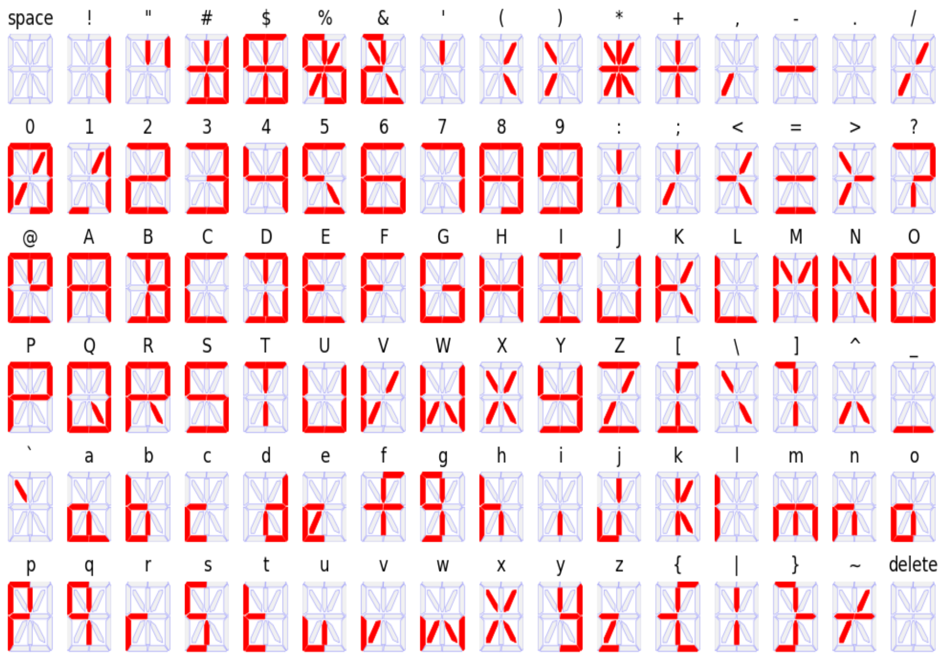
Figure S2: Original and Fake logic circuits for S-segment of sixteen-segment display. (a) represent the original circuit diagram, the generated fake version of original circuits are depicted in Sub-figures (b), (c), (d), (e), and (f)



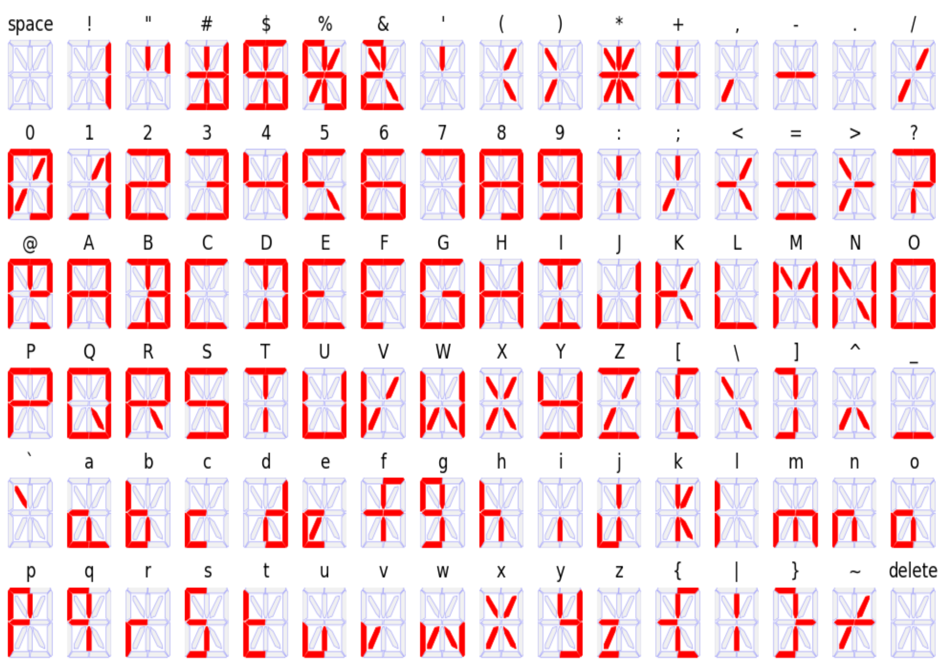
(a) Original: Sixteen-segment display output (all characters, digits and symbols are represented with correct segments)



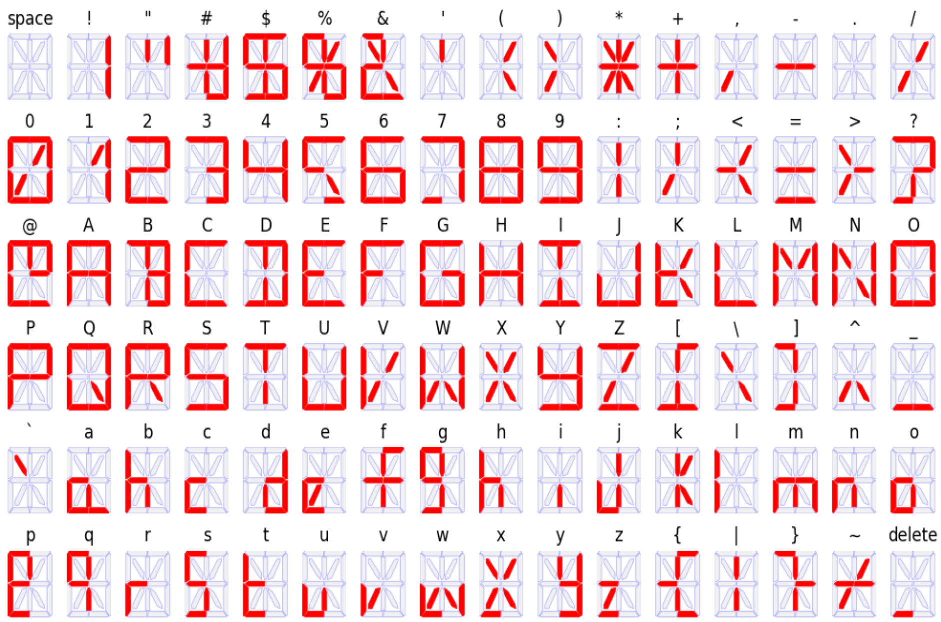
(b) Fake 1: output differs by a set of characters (D, L, and T), symbols (@, \), and digit (2).



(c) Fake 2: output differs with the digits (0,1 and 8) and symbol ([).



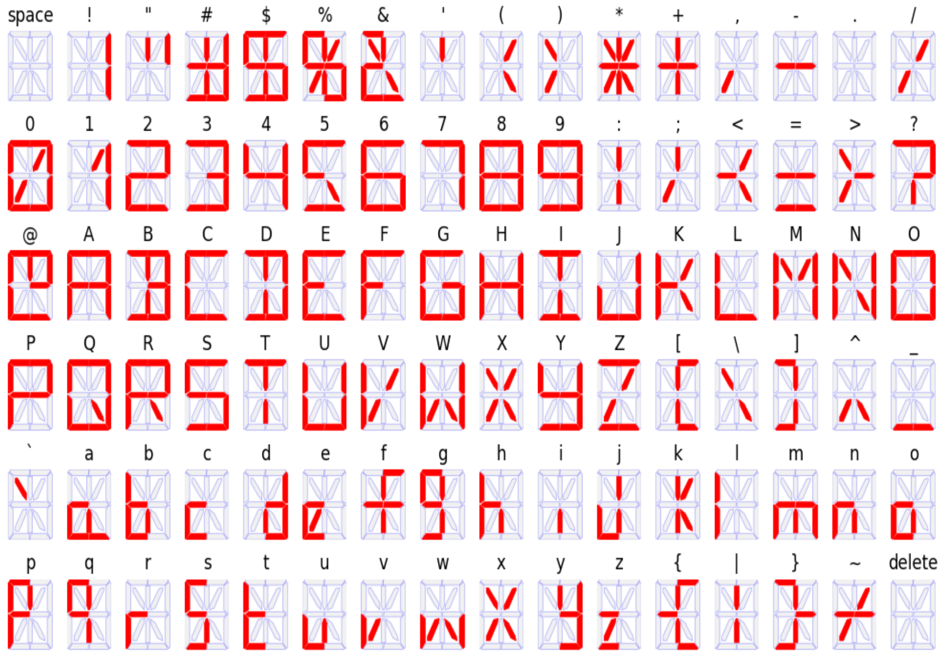
(d) Fake 3: output differs with the digits (0,1 and 8) and symbol(@).



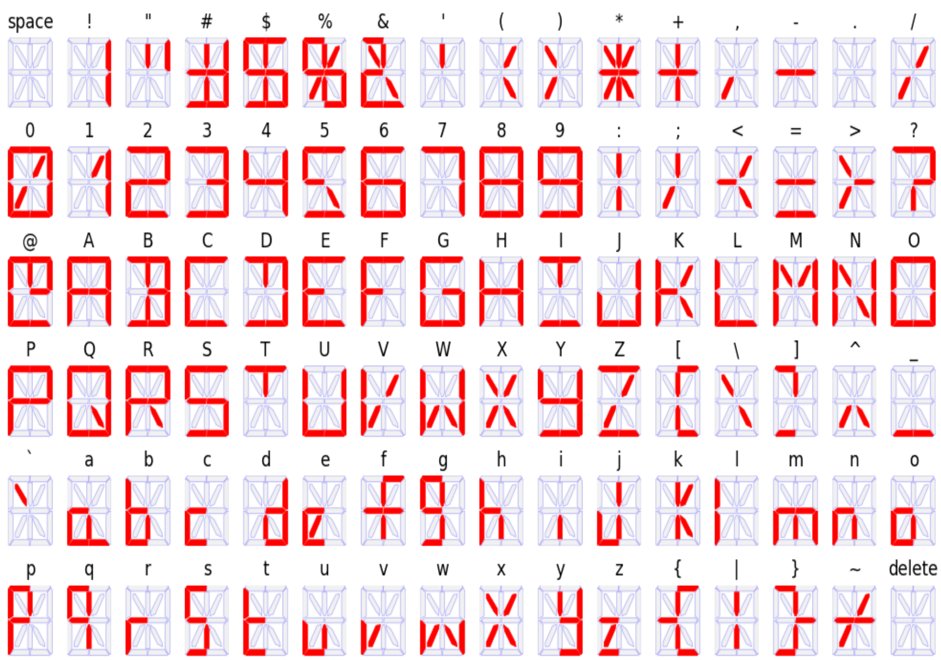
(e) Fake 4: output differs with digits 5 and 7, characters (B, K, b, w, and x), and special symbols (?, [, #, } and delete).

(f) Fake 5: output differs in output for digits (2 and 6), symbols(@ and <), and characters (P and T).

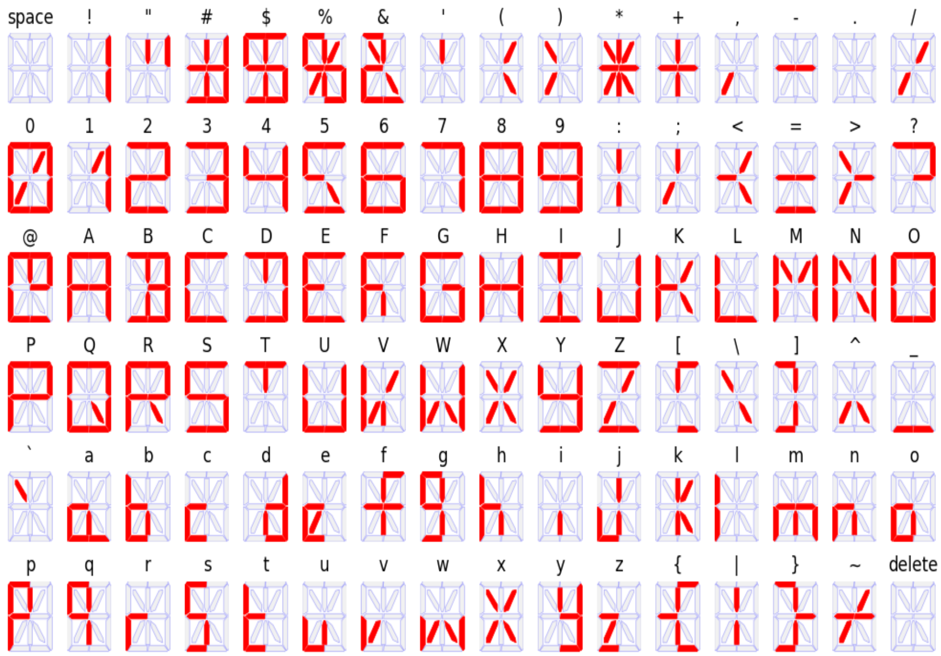
Figure S3: Sixteen-Segment Display Output for Original and Fake Logic Circuits after manipulation in f-segment: It is apparent that there is a significant difference in the output of fake logic circuits obtained through the experiment.



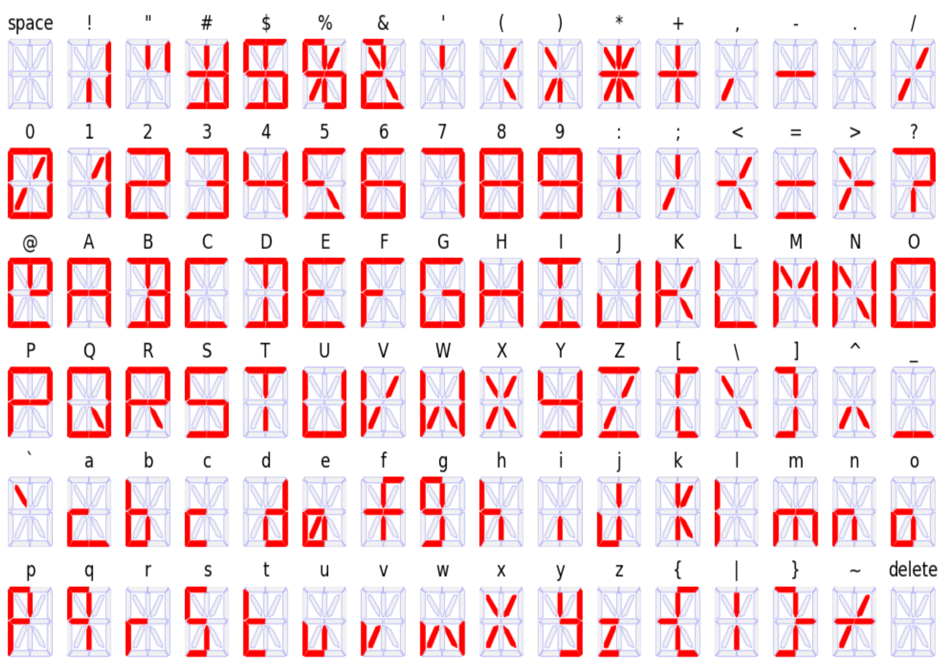
(a) Original: Sixteen-segment display output (all characters, digits and symbols are represented with correct segments)



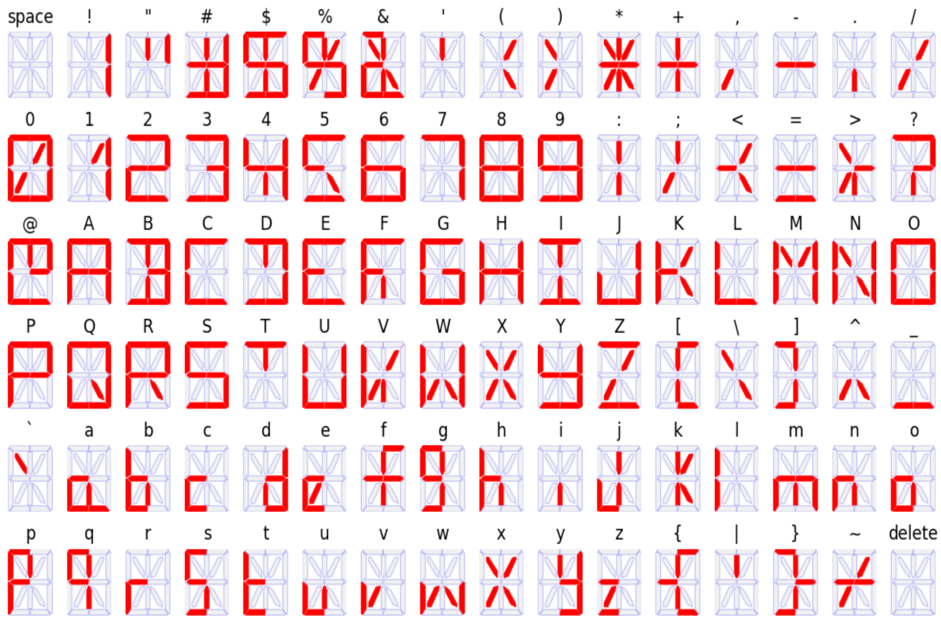
(b) Fake 1: incorrect output for the characters (D, I, and T) and symbol (]).



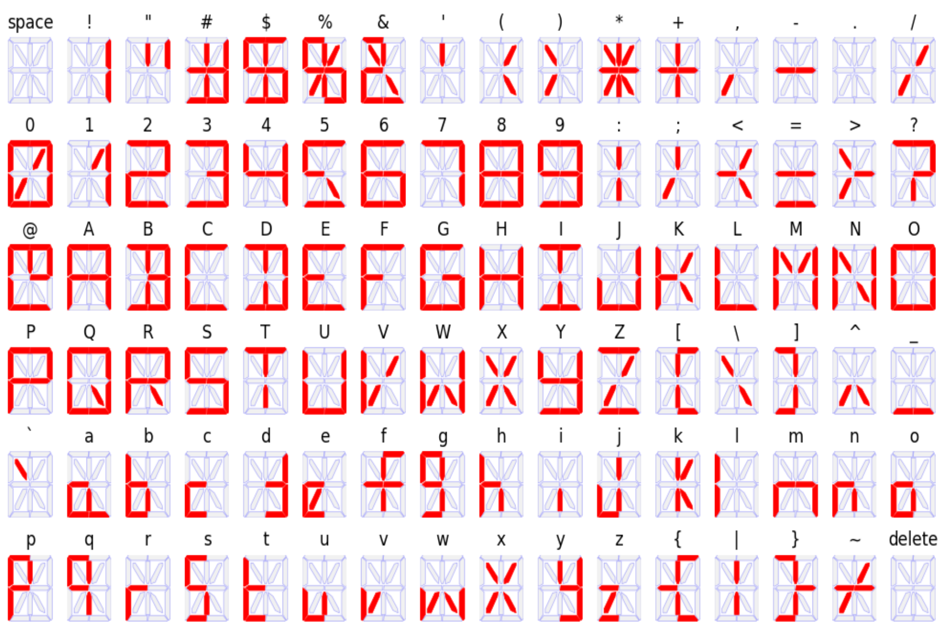
(c) Fake 2: incorrect outputs are ?, D, T, V, and [.



(d) Fake 3: output differs with symbol ")" and characters a, and e.

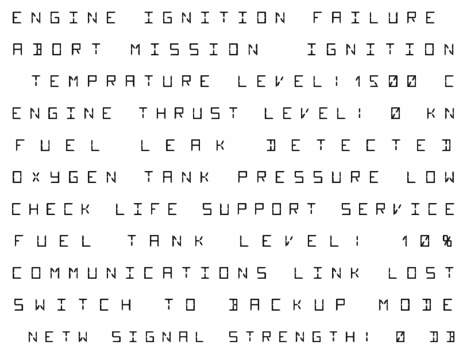


(e) Fake 4: output differs with &, 4, >, D, F, T, and V.

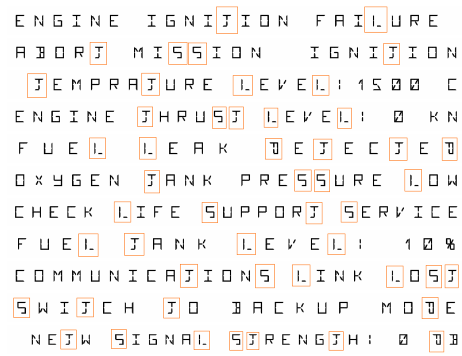


(f) Fake 5: output differs with character d and m.

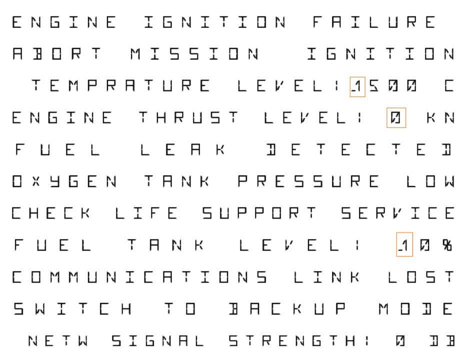
Figure S4: Sixteen-Segment Display Output for Original and Fake Logic Circuits after manipulation in s-segment: It is evident that the output of display for fake circuits are significantly different from the output of the display obtained using Original circuit.



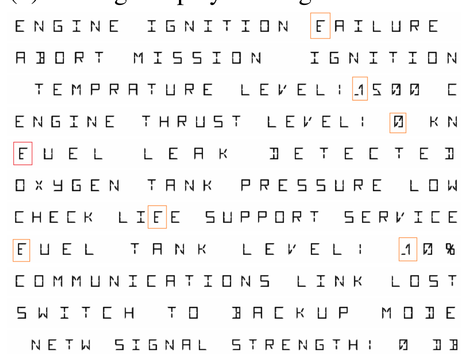
(a) Message displayed using original circuits



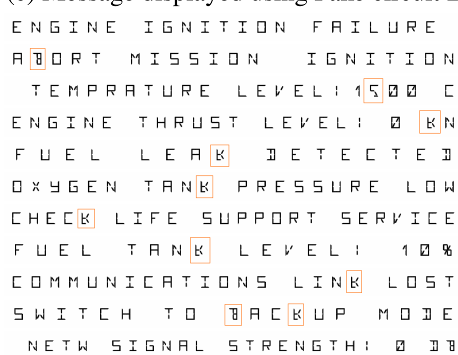
(b) Message displayed using Fake circuit 1



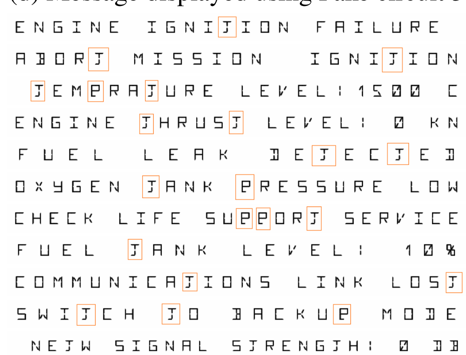
(c) Message displayed using Fake circuit 2



(d) Message displayed using Fake circuit 3

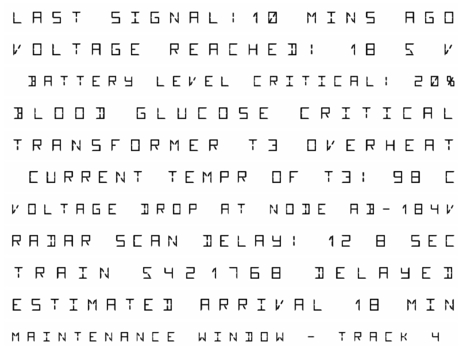


(e) Message displayed using Fake circuit 4

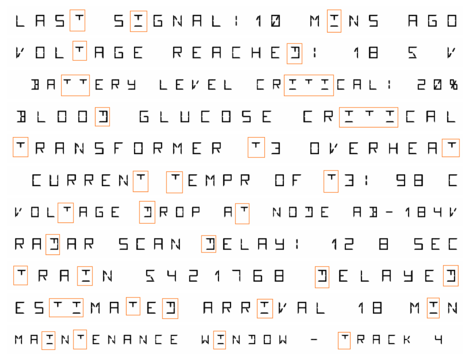


(f) Message displayed using Fake circuit 5

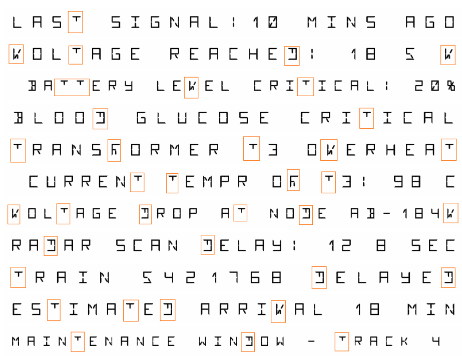
Figure S5: Generated alert messages using concatenated original and fake sixteen-segment displays after manipulation in f-segment (the highlighted characters represent segments where the fake circuit introduces subtle structural modifications, resulting in distorted characters digits or symbols).



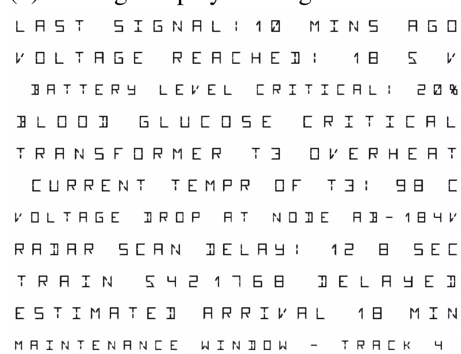
(a) Message displayed using original circuits



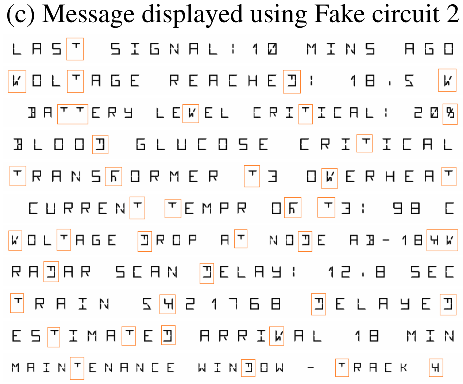
(b) Message displayed using Fake circuit 1



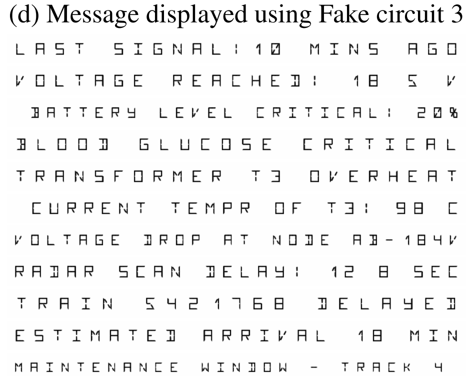
(c) Message displayed using Fake circuit 2



(d) Message displayed using Fake circuit 3



(e) Message displayed using Fake circuit 4



(f) Message displayed using Fake circuit 5

Figure S6: Generated alert messages using concatenated original and fake sixteen-segment displays after manipulation in s-segment (the highlighted characters represent segments where the fake circuit introduces subtle structural modifications, resulting in distorted characters digits or symbols).